

Improving Localization of Far-Sidelobe FRBs Detected by CHIME/FRB

Isabella Rivera¹, Nina Gusinskaia², Juan Mena-Parra^{1,3}, CHIME/FRB Sidelobe Working Group

1. David A. Dunlap Department of Astronomy & Astrophysics, University of Toronto, 2. ASTRON, the Netherlands Institute for Radio Astronomy
3. Dunlap Institute for Astronomy & Astrophysics, University of Toronto



Introduction

Fast radio bursts (FRBs) are millisecond duration radio pulses from outside the Milky Way whose physical origins remain unknown¹. A key step toward understanding FRB progenitors is **precise localization**, which allows bursts to be connected to their **host galaxies** and local environments.

CHIME/FRB, the fast radio burst project on the Canadian Hydrogen Intensity Mapping Experiment, is the world's most prolific FRB detector, having discovered over 4,500 bursts to date². CHIME detects FRBs not only through its **main beam**, but also through **lower-sensitivity regions called far-sidelobes** (defined as 5° off-meridian in E - W direction). These detections are rare, but especially valuable: because only very bright bursts can be detected in the sidelobes, they may indicate a **nearby FRB population**³.

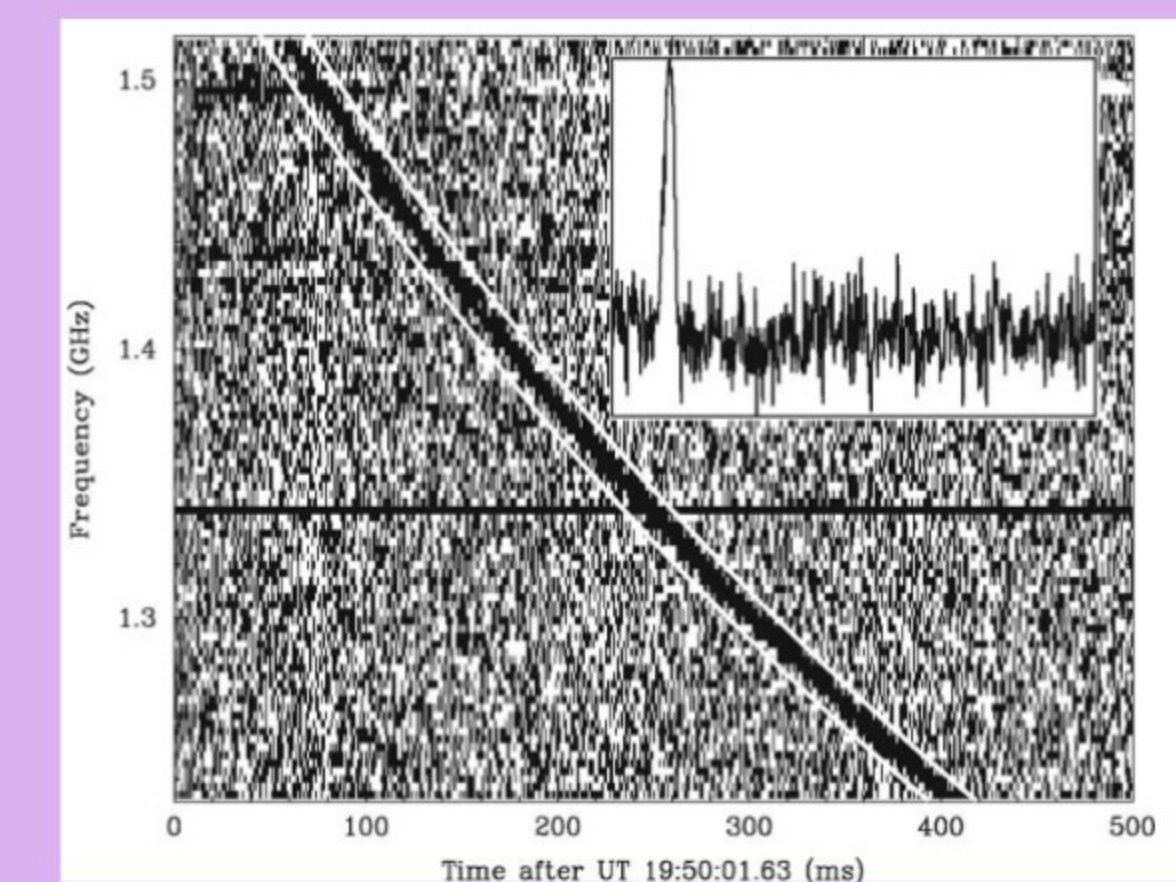


Figure 1: Dispersed burst of the first detected FRB.

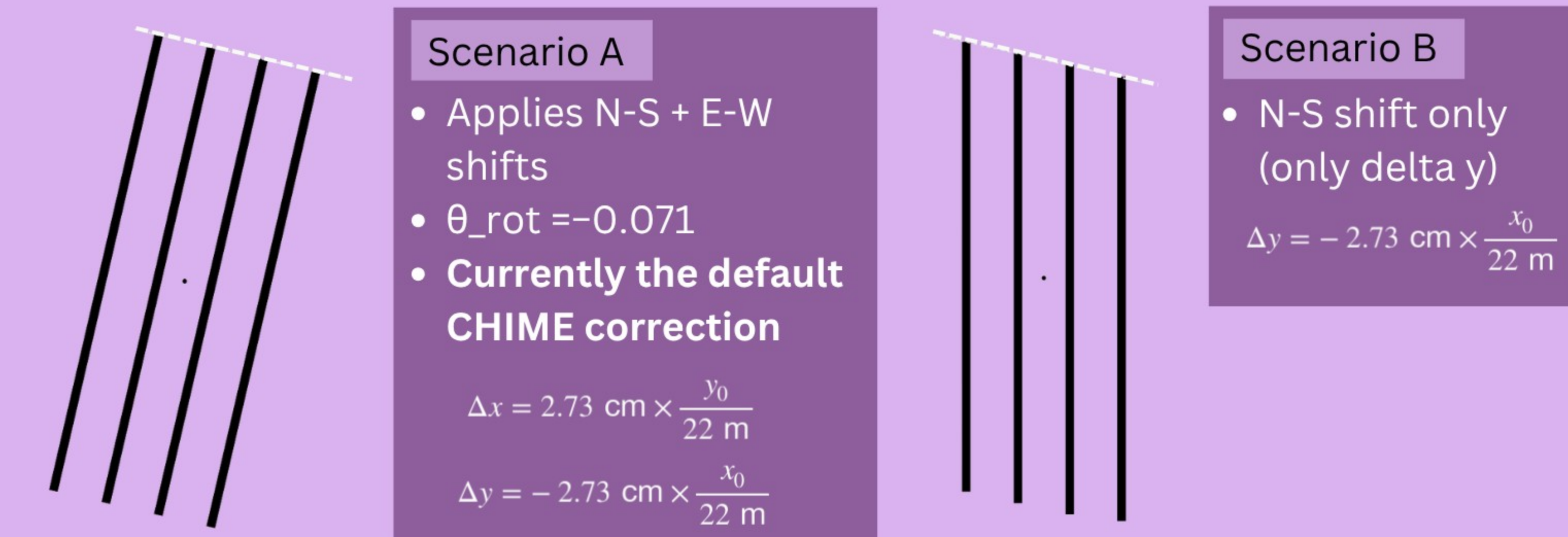


Figure 2: The CHIME radio telescope, a four cylinder transit interferometer located in British Columbia, Canada.

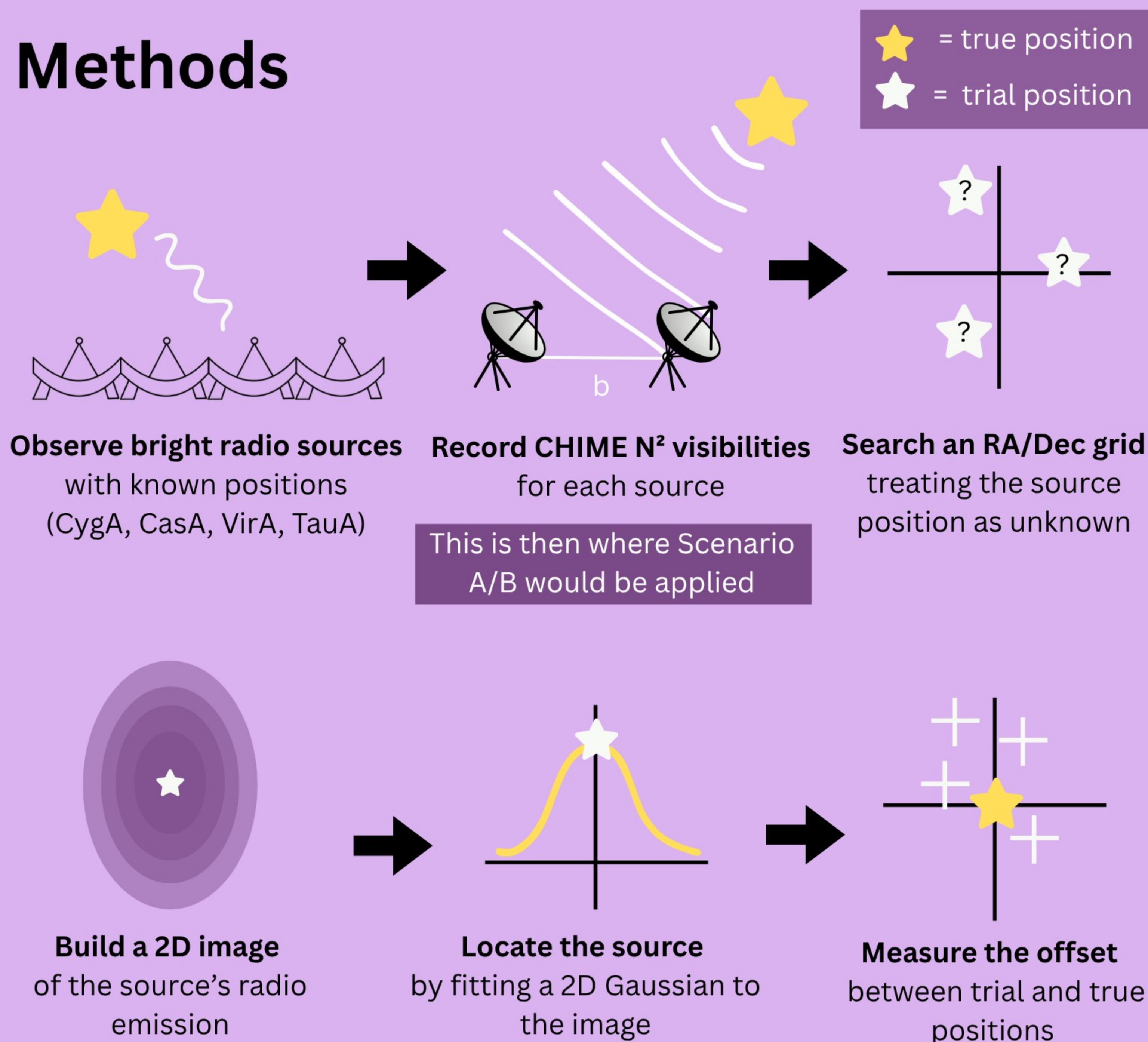
The Challenge

However, **far-sidelobe localization is challenging**. Small geometric systematics in the CHIME cylinders can shift the reconstructed sky position, limiting the reliability of host galaxy searches. This project tests CHIME geometric correction models using bright steady radio sources to **improve far-sidelobe FRB localization**.

Scenario A vs Scenario B



Methods



Results

After localizing each bright radio source, we compared recovered positions before and after each correction. We compared higher-dec sources (Cas A, Cyg A) and lower-dec sources (Tau A, Vir A) to test whether the corrections perform differently across the sky.

Scenario A is not a global improvement

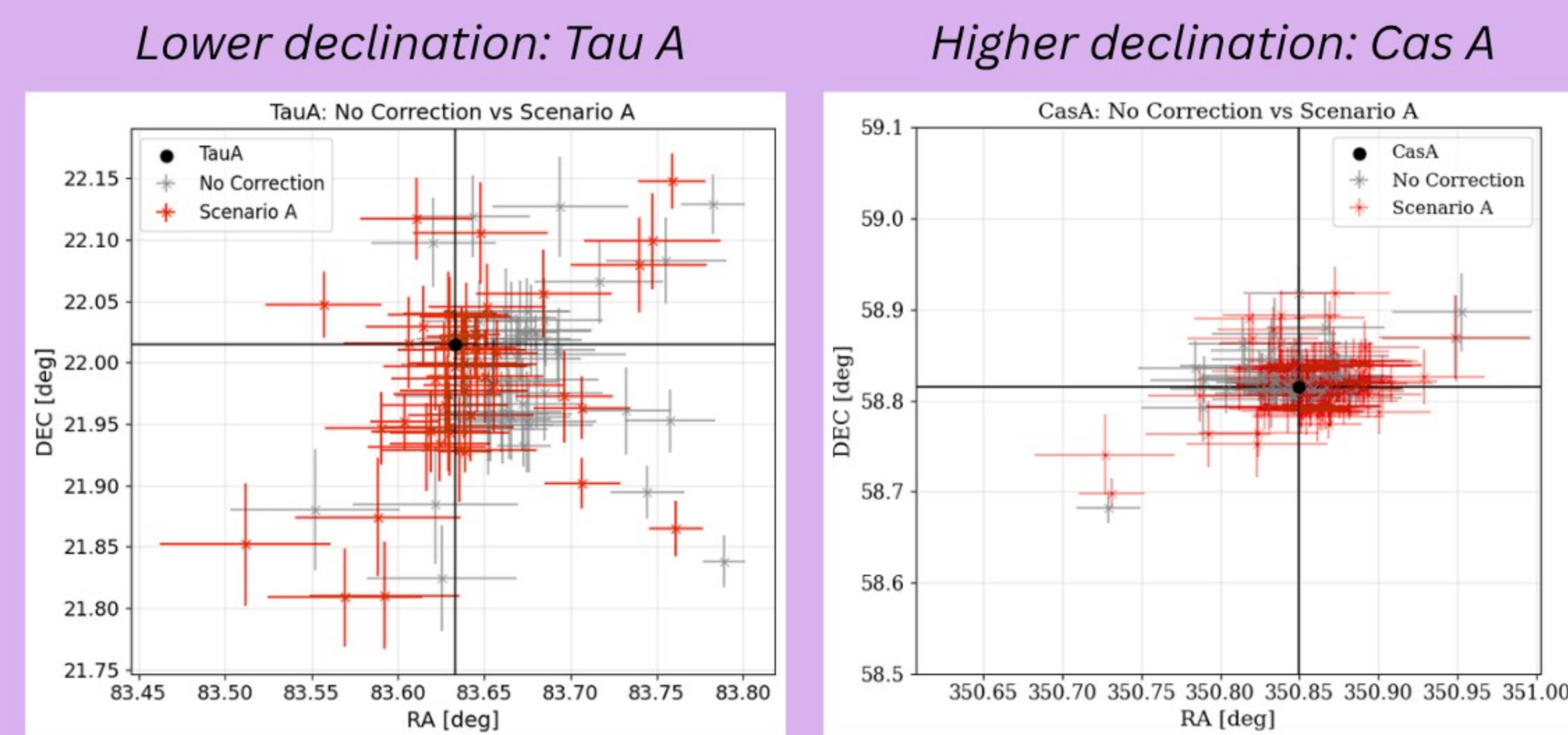


Figure 3: Tau A positions before and after Scenario A.

Figure 4: Cas A positions before and after Scenario A.

Total RMS (arcmin)	
Tau A	
No Correction	5.32'
Scenario A	5.18'
↑ 3% better	
Cas A	
No Correction	1.83'
Scenario A	2.28'
↓ 25% worse	

Scenario A improves low-dec localization, but worsens high-dec localization.

Scenario B provides the most consistent improvement

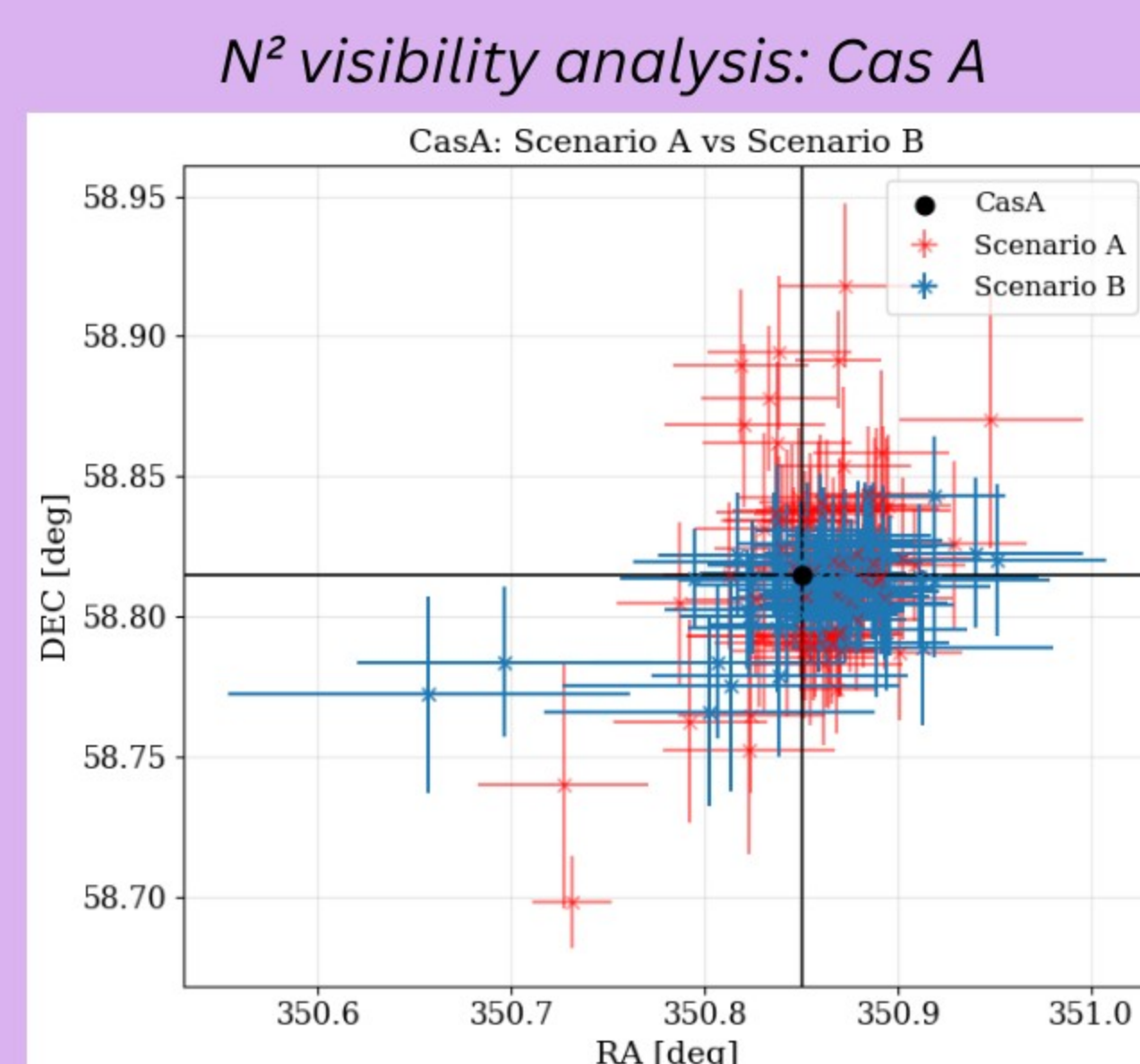
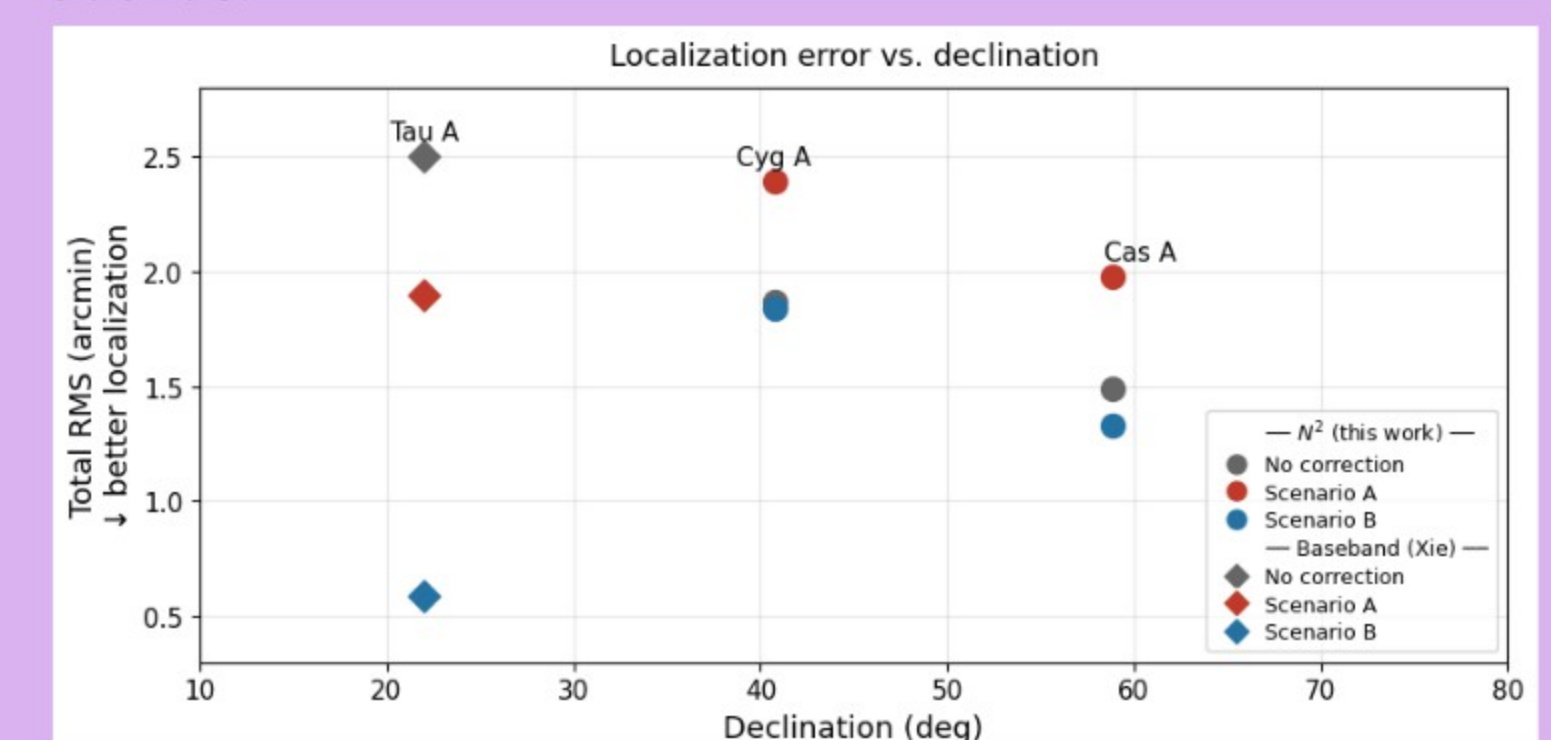


Figure 5: Cas A N² visibility localization under Scenarios A and B.

Figure 6: Localization error vs. declination for each source.



Independent analysis of steady and transient sources show consistent qualitative behavior. **Scenario B provides the most consistent localization improvement across source declinations**⁵.

Baseband analysis: Crab (Tau A)

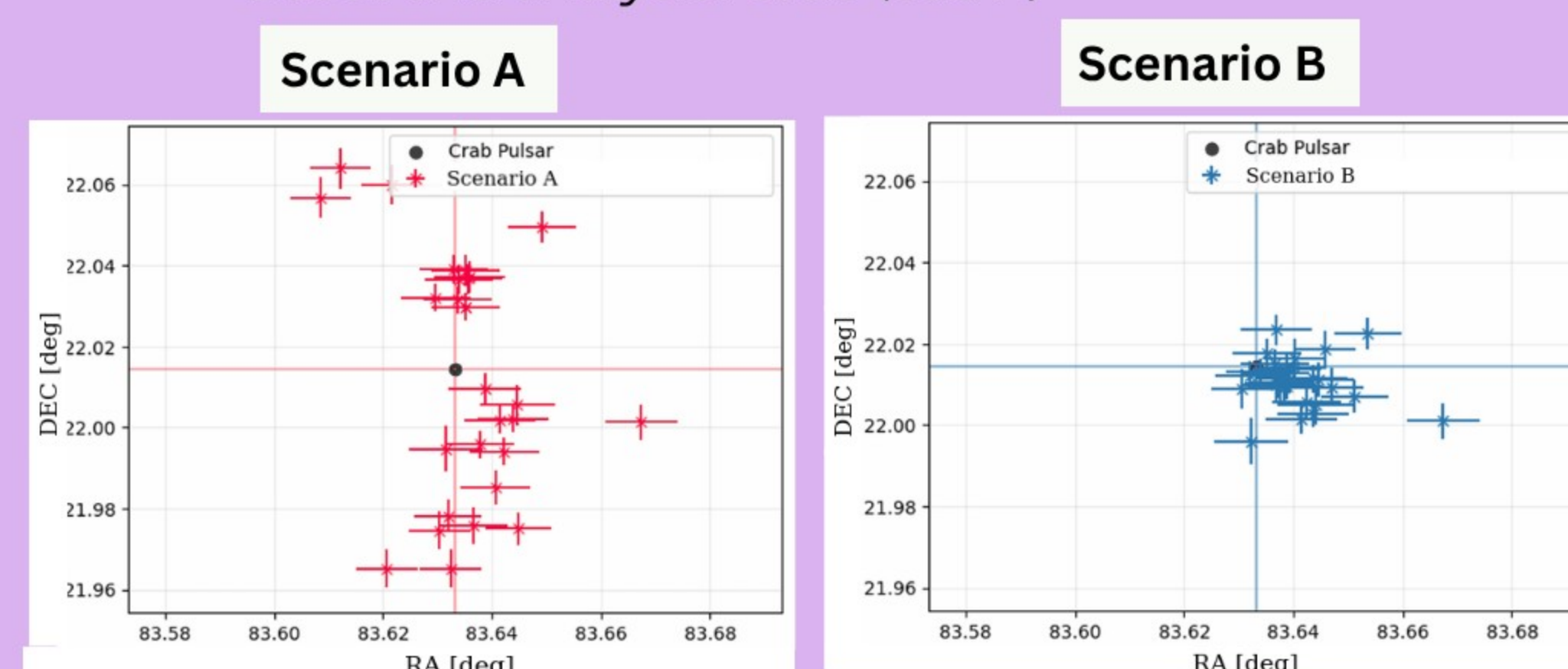


Figure 7: Crab (Tau A) baseband localization applying Scenario A.

Figure 8: Crab (Tau A) baseband localization applying Scenario B.

Total RMS (arcmin)	
Cas A (N² vis)	
Scenario A	2.28'
Scenario B	1.70'
↑ 25% better	
Crab (BB)	
Scenario A	1.90'
Scenario B	0.59'
↑ 69% better	

Baseband localization analysis performed by Chessy Xie.

Host Associations

By combining CHIME/FRB localizations of 21 far-sidelobe events with a probabilistic framework for associating radio transients to host galaxies⁴, we identified 4 promising candidates requiring spectroscopic follow-up.

FRB20211217B has a host association probability of **98.7%** with galaxy IC 520.

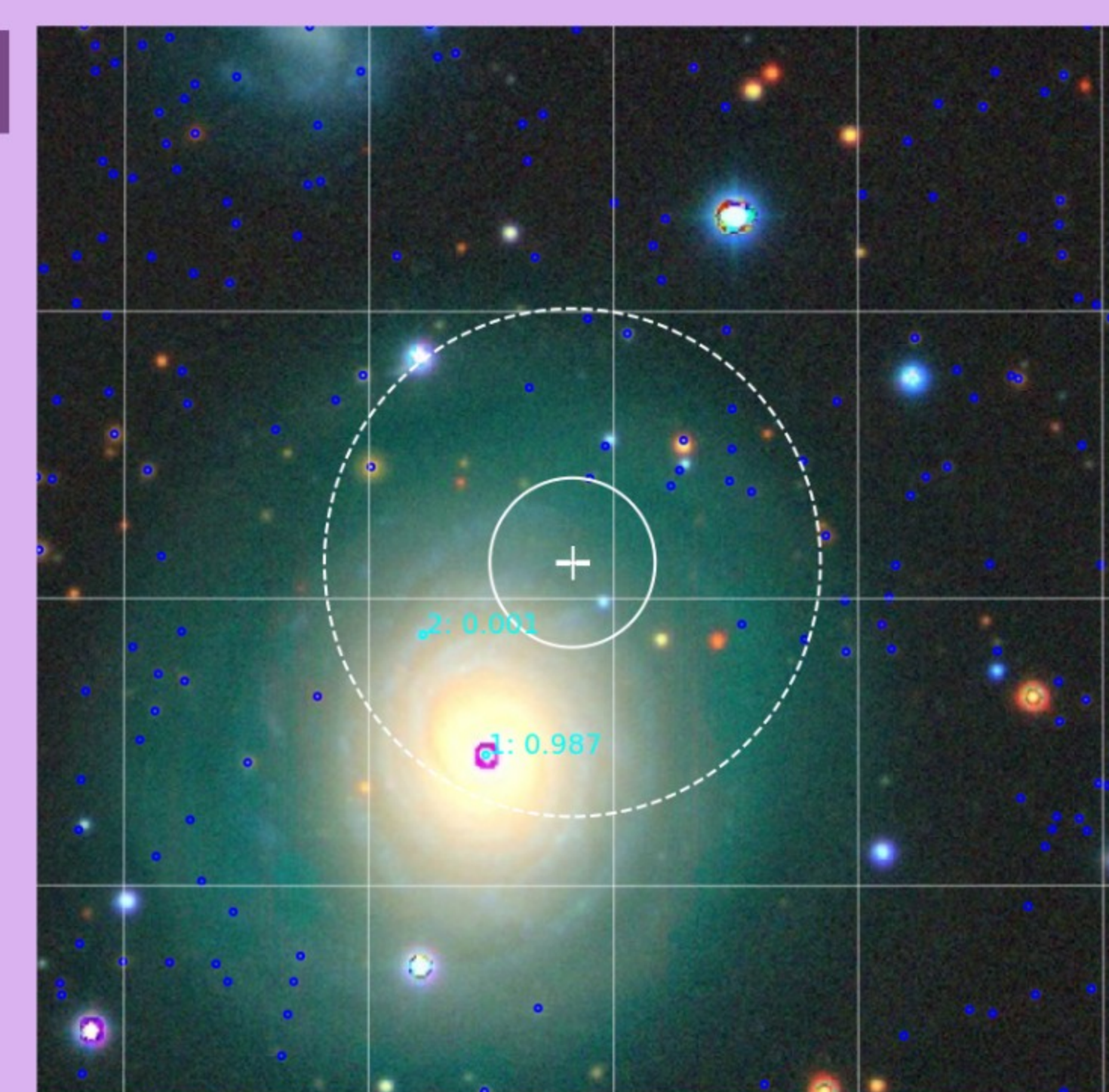


Figure 9: FRB20211217B was found to have a potential host association.

Next Steps

Apply Scenario B to future far-sidelobe FRB detections to improve their localization accuracy and strengthen possible host galaxy associations.

- Strengthen host galaxy associations
- Prioritize spectroscopic follow-up targets
- Better characterize the nearby FRB population

